

Appendix B

Number Systems on the Computer

B.1 Binary Numbers

Humans like to use the *decimal number* system for representing numbers. Decimal numbers are *base 10* meaning that a decimal number consists of a sequence of digits separated by a *decimal point*, where each *digit* can have values $d \in \{0, 1, 2, \dots, 9\}$ and the weight of each digit is proportional to its place in the sequence of digits with respect to the decimal point, i.e., the number $357.6 = 3 \cdot 10^2 + 5 \cdot 10^1 + 7 \cdot 10^0 + 6 \cdot 10^{-1}$, or in general, for a number consisting of digits d_i with $n + 1$ and m digits to the left and right of the decimal point, the value v is calculated as:

$$v = \sum_{i=-m}^n d_i 10^i. \quad (\text{B.1})$$

The basic unit of information in almost all computers is the binary digit, or *bit* for short. A *binary number* consists of a sequence of binary digits separated by a decimal point, where each digit can have values $b \in \{0, 1\}$, and the base is 2. The general equation is,

$$v = \sum_{i=-m}^n b_i 2^i, \quad (\text{B.2})$$

and examples are $1011.1_2 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 + 1 \cdot 2^{-1} = 11.5$. Notice that we use subscript 2 to denote a binary number, while no subscript is used for decimal numbers. The left-most bit is called the *most significant bit*, and the right-most bit is called the *least significant bit*. Due to typical organisation of computer memory, 8 binary digits is called a *byte*, and the term *word* is not universally defined but typically related to the computer architecture, a program is running on, such as 32 or 64 bits.

Other number systems are often used, e.g., *octal numbers*, which are base 8 numbers and have digits $o \in \{0, 1, \dots, 7\}$. Octals are useful short-hand for binary, since 3 binary digits map to the set of octal digits. Likewise, *hexadecimal numbers* are base 16 with digits $h \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, a, b, c, d, e, f\}$, such that $a_{16} = 10$, $b_{16} = 11$ and so on. Hexadecimals are convenient, since 4 binary digits map directly to the set of hexadecimal digits. Thus $367 = 101101111_2 = 557_8 = 16f_{16}$. A list of the integers 0–63 in various bases is given in Table B.1.

Dec	Bin	Oct	Hex	Dec	Bin	Oct	Hex
0	0	0	0	32	100000	40	20
1	1	1	1	33	100001	41	21
2	10	2	2	34	100010	42	22
3	11	3	3	35	100011	43	23
4	100	4	4	36	100100	44	24
5	101	5	5	37	100101	45	25
6	110	6	6	38	100110	46	26
7	111	7	7	39	100111	47	27
8	1000	10	8	40	101000	50	28
9	1001	11	9	41	101001	51	29
10	1010	12	a	42	101010	52	2a
11	1011	13	b	43	101011	53	2b
12	1100	14	c	44	101100	54	2c
13	1101	15	d	45	101101	55	2d
14	1110	16	e	46	101110	56	2e
15	1111	17	f	47	101111	57	2f
16	10000	20	10	48	110000	60	30
17	10001	21	11	49	110001	61	31
18	10010	22	12	50	110010	62	32
19	10011	23	13	51	110011	63	33
20	10100	24	14	52	110100	64	34
21	10101	25	15	53	110101	65	35
22	10110	26	16	54	110110	66	36
23	10111	27	17	55	110111	67	37
24	11000	30	18	56	111000	70	38
25	11001	31	19	57	111001	71	39
26	11010	32	1a	58	111010	72	3a
27	11011	33	1b	59	111011	73	3b
28	11100	34	1c	60	111100	74	3c
29	11101	35	1d	61	111101	75	3d
30	11110	36	1e	62	111110	76	3e
31	11111	37	1f	63	111111	77	3f

Table B.1 A list of the integers 0–63 in decimal, binary, octal, and hexadecimal.

B.2 IEEE 754 Floating Point Standard

The set of real numbers, also called *reals*, includes all fractions and irrational numbers. It is infinite in size both in the sense that there is no largest nor smallest number, and that between any 2 given numbers there are infinitely many numbers. Reals are widely used for calculation, but since any computer only has finite memory, there are infinitely many numbers which cannot be represent on a computer. Hence, any computation performed on a computer with reals must rely on approximations. *IEEE 754 double precision floating-point format (binary64)*, known as a *double*, is a standard for representing an approximation of reals using 64 bits. These bits are divided into 3 parts: sign, exponent and fraction,

$$s e_1 e_2 \dots e_{11} m_1 m_2 \dots m_{52},$$

where s , e_i , and m_j are binary digits. The bits are converted to a number using the equation by first calculating the exponent e and the mantissa m ,

$$e = \sum_{i=1}^{11} e_i 2^{11-i}, \quad (\text{B.3})$$

$$m = \sum_{j=1}^{52} m_j 2^{-j}. \quad (\text{B.4})$$

I.e., the exponent is an integer, where $0 \leq e < 2^{11}$, and the mantissa is a rational, where $0 \leq m < 1$. For most combinations of e and m , the real number v is calculated as,

$$v = (-1)^s (1 + m) 2^{e-1023} \quad (\text{B.5})$$

with the exceptions that

	$m = 0$	$m \neq 0$
$e = 0$	$v = (-1)^s 0$ (signed zero)	$v = (-1)^s m 2^{1-1023}$ (subnormals)
$e = 2^{11} - 1$	$v = (-1)^s \infty$	$v = (-1)^s \text{NaN}$ (not-a-number)

where $e = 2^{11} - 1 = 11111111111_2 = 2047$. The largest and smallest number that is not infinity is thus

$$e = 2^{11} - 2 = 2046, \quad (\text{B.6})$$

$$m = \sum_{j=1}^{52} 2^{-j} = 1 - 2^{-52} \simeq 1, \quad (\text{B.7})$$

$$v_{\max} = \pm (2 - 2^{-52}) 2^{1023} \simeq \pm 2^{1024} \simeq \pm 10^{308}. \quad (\text{B.8})$$

The density of numbers varies in such a way that when $e - 1023 = 52$, then

$$v = (-1)^s \left(1 + \sum_{j=1}^{52} m_j 2^{-j} \right) 2^{52} \quad (\text{B.9})$$

$$= \pm \left(2^{52} + \sum_{j=1}^{52} m_j 2^{-j} 2^{52} \right) \quad (\text{B.10})$$

$$= \pm \left(2^{52} + \sum_{j=1}^{52} m_j 2^{52-j} \right) \quad (\text{B.11})$$

$$\stackrel{k=52-j}{=} \pm \left(2^{52} + \sum_{k=51}^0 m_{52-k} 2^k \right), \quad (\text{B.12})$$

which are all integers in the range $2^{52} \leq |v| < 2^{53}$. When $e - 1023 = 53$, then the same calculation gives

$$v \stackrel{k=53-j}{=} \pm \left(2^{53} + \sum_{k=52}^1 m_{53-k} 2^k \right), \quad (\text{B.13})$$

which are every second integer in the range $2^{53} \leq |v| < 2^{54}$, and so on for larger values of e . When $e - 1023 = 51$, the same calculation gives,

$$v \stackrel{k=51-j}{=} \pm \left(2^{51} + \sum_{k=50}^{-1} m_{51-k} 2^k \right), \quad (\text{B.14})$$

which is a distance between numbers of $1/2$ in the range $2^{51} \leq |v| < 2^{52}$, and so on for smaller values of e . Thus we may conclude that the distance between numbers in the interval $2^n \leq |v| < 2^{n+1}$ is 2^{n-52} , for $-1022 = 1 - 1023 \leq n < 2046 - 1023 = 1023$. For subnormals, the distance between numbers is

$$v = (-1)^s \left(\sum_{j=1}^{52} m_j 2^{-j} \right) 2^{-1022} \quad (\text{B.15})$$

$$= \pm \left(\sum_{j=1}^{52} m_j 2^{-j} 2^{-1022} \right) \quad (\text{B.16})$$

$$= \pm \left(\sum_{j=1}^{52} m_j 2^{-j-1022} \right) \quad (\text{B.17})$$

$$\stackrel{k=-j-1022}{=} \pm \left(\sum_{j=-1023}^{-1074} m_{-k-1022} 2^k \right), \quad (\text{B.18})$$

which gives a distance between numbers of $2^{-1074} \simeq 10^{-323}$ in the range $0 < |v| < 2^{-1022} \simeq 10^{-308}$.